Linear Algebra — Mid-Term Model Test 3

Covers Sections 1.1–1.3 and 2.1–2.6 · Level: hard

Instructions. Time allowed: 150 minutes. Calculators are *not* allowed. Show all working. Total: 100 points. Difficulty: hard (beyond the sample test; includes parameters and short proofs).

1 Vectors

(a) Prove that for any vectors u, v: $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$. **[6 pts]**
(b) Find all vectors $w \in \mathbb{R}^3$ that are perpendicular to both $(1,1,1)$ and $(1,2,3)$. **[6 pts]**
(c) In a unit cube, find the cosine of the angle between the main diagonal $(1,1,1)$ and (i) an edge $(1,0,0)$, (ii) a face diagonal $(1,1,0)$. **[6 pts]**

2 Matrix powers and polynomials

(1) Let $A = \begin{pmatrix} 3 & 1 & 2 \\ 6 & 2 & 4 \\ -3 & -1 & -2 \end{pmatrix}$.
(a) Compute A^2 and use it to find A^{10}. **[6 pts]**
(b) Find $(I + A)^5$. **[8 pts]**
(2) Let $J = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$.
(a) Compute J^{10}. **[6 pts]**
(b) Compute $f(J)$ for $f(x) = x^4 - 2x^2 + 3$. **[6 pts]**

3 Elimination with a parameter

For each real λ, decide whether elimination succeeds, and find all solutions of **[12 pts]**

$$\begin{cases} \lambda x_1 + x_2 + x_3 = 1, \\ x_1 + \lambda x_2 + x_3 = \lambda, \\ x_1 + x_2 + \lambda x_3 = \lambda^2. \end{cases}$$

4 Inverses

(1) Calculate the inverse of **[12 pts]**

$$P = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 3 & -2 & -1 & 3 \\ 3 & -1 & 1 & 2 \\ 1 & -1 & -2 & 3 \end{pmatrix}.$$

(2) Find the matrix X satisfying $AX = A + 2X$, where $A = \begin{pmatrix} 3 & -1 & 0 \\ 0 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$. **[8 pts]**

5 $A = LU$ with parameters

(a) Find L and U for $A = \begin{pmatrix} a & a & a \\ a & b & b \\ a & b & c \end{pmatrix}$. For which a, b, c does A have three pivots (so that $A = LU$ with U invertible)? **[8 pts]**

(b) Take $a = 1$, $b = 2$, $c = 4$ and solve $Ax = (1, 3, 7)^T$ using L and U. **[6 pts]**

6 Proofs

(a) Let A be a square matrix with $A^2 - A - 2I = 0$. Prove that A and $A - I$ are invertible and find their inverses in terms of A. **[5 pts]**

(b) Let N be a square matrix with $N^k = 0$ for some integer $k \geq 1$. Prove that $I - N$ is invertible. **[5 pts]**